

CHAPTER-1

INTODUCTION

ORIGIN OF FINANCE:

Finance originated around 3000 BCE in ancient Mesopotamia, where temples and places served as early banks for storing valuables, leading to the development of credit, loans, by 1800 BCE formal debt regulation in the code of Hammurabi. It evolved from barter systems to metallic money and banking, formalizing in medieval Italy and establishing modern theory by the 1950s.

1. Ancient beginnings

- Finance began with the barter system, where goods were exchanged directly.
- Around 3000 BCE in Mesopotamia, people started recording transactions on clay tablets.
- Temples and palaces acted like early financial institutions, storing wealth and lending grain or silver.

2. Introduction of money

- The first metal coins were introduced in Lydia around the 7th century BCE
- Money replaced barter, making trade easier and more efficient.
- This led to the growth of markets and economic activity.

3. Birth of banking

- Banking systems emerged in ancient Greece and Rome.
- Modern banking developed in medieval Italy, especially in cities like Florence.
- The Medici family played a major role in developing banking practices like loans, deposits, and bookkeeping.

4. Development of financial markets

- The first stock exchange was established in Amsterdam in the 17th century.
- Companies began issuing shares to raise capital.
- Financial markets expanded globally, enabling investment and economic growth

Modern Finance

Finance today includes banking, investments, insurance, and financial technology (fintech).

Global institutions, stock markets, and digital platforms dominate.

Concepts like corporate finance, public finance, and persona finance evolved.

INTRODUCTION TO FINANCE

Finance is an important field that contributes significantly to the functioning of moder economies. It includes a wide range of operations involving the management of money and other financial resources. Finance is a vital for making intelligent budgeting, investing, saving, and spending decisions at all levels, from individual homes to organizations and governments. At its foundation, finance is about knowing how money works and how to use it to achieve specific objectives. This includes raising cash in a variety of ways, such as issuing stocks on bonds, managing assets to achieve maximum return, and assessing risks to reduce potential losses.

1.2 DEFINITION OF FINANCE:

Finance is the science and practice of managing money and other financial assets. It entails the study, creation, and administration of funds, including bow money is raised, invested, and spent by individuals, businesses, and governments, Finance's fundamental purpose is to optimize resource allocation, control risks, and maximize investment value.

1.3 MEANING OF FINANCE:

Finance is the science and art of managing money and financial resources effectively, and the activity of earning, spending, saving, investing, and managing money.

1.4 SCOPE OF FINANCE:



- **Strategy and Planning** - A strategy is a comprehensive plan or method used to accomplish a certain goal or group of goals. It entails choosing how to compete in the market, where to put resources, and what steps to follow in order to accomplish long-term goals. Planning is the process of establishing objectives and deciding the actions required to accomplish them.
- **Treasury Management** - Treasury management is the oversight and control of an organization's financial resources, usually by its treasury department or function. The fundamental purpose of treasury management is to guarantee that the business has enough liquidity to meet its financial obligations and improve its overall financial performance.
- **Investor Relations** -Investor Relations (IR) is the strategic management department within a corporation that communicates with investors, analysts, shareholders, and other stakeholders. The major purpose of Investor Relations is to foster strong connections and open communication between the company and the financial community.

- **Investment Banking-** Investment banking is a type of banking that specializes in providing financial advice services, raising capital, and conducting mergers and acquisitions (M&A) for firms, governments, and other organizations.
- **Investment Analysis** - Investment analysis aims to help investors make informed decisions that optimize the risk-return trade-off while also aligning with their financial goals and preferences.
- **Investment Management-** Investment management, often known as asset management, is the professional management of various securities and assets (such as stocks, bonds, real estate, commodities, and other investments) on behalf of individuals or institutions.
- **Financial Planning** - Financial planning is a complete process that assists people, families, and organizations in assessing their financial condition, setting financial goals, and developing a strategy to accomplish those goals.
- **Real Estate** - The term "real estate" refers to property that includes land, buildings, natural resources, and other improvements attached or located on it. It is a physical asset class that includes a variety of residential, commercial, industrial, and agricultural properties.
- **Insurance-** Insurance is a financial agreement that protects against the possibility of financial loss. The risk of potential losses is transferred from an individual or corporation (the insured) to an insurance firm (the insurer) in exchange for the payment of a premium.
- **Commercial Banking** - Commercial banking encompasses the banking activities and services offered by financial institutions (banks) to enterprises, corporations, and government agencies.

1.5 ADVANTAGES OF FINANCE:

a) Ensures Availability of Funds

Finance helps in raising and allocating capital as per the needs of the business or individual ensures liquidity and smooth day-to-day operations.

b) Supports Business Growth and Expansion

Adequate financing enables companies to invest in new projects, expand operations, or enter new markets. Helps in technology upgrades, research and development, and hiring skilled employees.

c) Aids in Decision-Making

Financial analysis helps in making informed decisions about investments, expenses, pricing and profitability, Tools like budgeting, forecasting, and cost-benefit analysis are finance based.

d) Increases Profitability

Effective financial management ensures optimal use of resources, minimizes costs, and enhances profits.

e) Risk Management

Finance involves tools and strategies (like insurance, diversification, hedging) that help in identifying, assessing, and minimizing risks.

f) Enhances Investor Confidence

Sound financial planning and transparent reporting boost investor trust and attract more investment Improves company's reputation and market valuation.

g) Facilitates Long-Term Planning

Finance helps individuals and organizations plan for the future—like retirement, expansion, or sustainability

h) Improves Operational Efficiency

With proper budgeting and fund allocation, businesses can reduce waste, control spending, and streamline operations.

i) Promotes Economic Development

At a national level, finance supports infrastructure development, industrialization, and social programs, driving economic growth.

1.6 DISADVANTAGES OF FINANCE:

1. Risk of Over-Borrowing

Excessive reliance on debt can lead to high interest costs and financial distress.

2. Cost of Financing Raising finance

whether through loans, equity, or issuing bonds—often involves high costs such as:

- Interest payments
- Administrative or underwriting fees
- Dividend obligations

3. Mismanagement or Misallocation of Funds

Improper financial planning can lead to wastage, unproductive investments, or cash flow problems.

4. Dependency on External Sources

Relying too much on external finance can make a business or individual vulnerable to creditors or investors.

5. Interest Rate Fluctuations

Loans and credit are sensitive to interest rate changes. Rising rates increase the cost of borrowing, reducing profitability.

6. Market Volatility and Investment Risks

Financial markets are unpredictable. Investments can lead to losses due to market crashes, inflation, or poor economic conditions.

7. Complexity and Regulation

Finance involves complex calculations, compliance, and legal regulations that require expertise.

8. Short-Term Focus

Companies may focus on short-term financial gains (like boosting EPS) rather than long-term sustainability, leading to unstable growth.

1.7 TYPES OF FINANCE

- ❖ PERSONAL FINANCE
- ❖ CORPORATE FINANCE
- ❖ PUBLIC FINANCE

- ❖ **Personal Finance** - Personal finance refers to managing an individual's or household's financial activities, such as budgeting, saving, investing, insurance, and retirement planning
- ❖ **Corporate Finance**-Corporate finance refers to a company's financial actions and decisions involving capital management, funding sources, investments, and financial strategies in order to maximize shareholder value and meet organizational goals.
- ❖ **Public Finance** - Public finance is the study of government spending, revenue generation, and debt management at all levels of government (national, state, and local) and public entities. It includes the policies, methods, and mechanisms by which governments collect revenue, allocate resources, and manage their finances in order to achieve economic and social goals.
- ❖ **Investment Finance** -Investment finance is the branch of finance that focuses on allocating financial resources to various investment possibilities with the goal of creating profits for investors. It includes examining investment possibilities, managing portfolios, and making strategic decisions to maximize investment performance while mitigating risk.
- ❖ **International Finance** - International finance is the branch of finance concerned with monetary interactions between countries, such as foreign investments and exchange rates It includes international financial management, investment, and trade activities.

1.8 ORIGIN OF LEVERAGE:

- In the 20th century, the term began to be used figuratively, particularly in finance, to refer to using borrowed money (debt) to increase the potential return on investment
- Financial leverage thus parallels the mechanical idea: using a small input (your own capital) to achieve a greater effect (larger returns) by adding external resources (borrowed funds).

1.9 DEFINITION OF LEVERAGE:

Leverage refers to the strategic use of fixed costs or borrowed funds to increase the potential return on investment. It magnifies both gains and losses, depending on how effectively the underlying resources are managed.

1.10 MEANING OF LEVERAGE:

Leverage is the strategic use of borrowed money (debt) to increase the potential return on investment, acquire assets, or fund business operations. It enables individuals and businesses to amplify their financial capacity and pursue growth opportunities beyond what their existing capital would allow. When an entity is described as highly leveraged, its debt is more significant than its equity, which gives an idea of its financial structure. Leverage can be a very effective tool but also poses substantial risks.

1.11 ADVANTAGES OF LEVERAGE

i. Increased Returns on Equity

- Leverage can amplify profits when sales and operations perform well.
- Especially true for financial leverage—using debt can boost Earnings Per Share (EPS) without issuing more equity.

ii. Tax Benefits

- Interest on debt is tax-deductible, which reduces taxable income.
- This makes debt a cheaper financing option compared to equity.

iii. Retain Ownership

- Using debt (financial leverage) allows a business to raise funds without diluting ownership.
- 4. Operational Efficiency
- High operating leverage encourages efficient use of fixed resources.
- Once fixed costs are covered, additional sales contribute significantly to profit.

iv. Strategic Growth

- Leverage enables companies to expand faster by funding projects or acquisitions with borrowed capital.

1.12 DISADVANTAGES OF LEVERAGE

1) Increased Financial Risk

- Higher leverage means higher fixed obligations (like interest payments).
- This increases the risk of bankruptcy if revenue falls.

2) Volatile Earnings

- Small changes in sales can cause large fluctuations in net income due to leverage.
- Makes earnings less predictable and riskier.

3) Debt Covenants & Restrictions

- Borrowing often comes with contractual limits imposed by lenders.
- May restrict future borrowing, dividends, or strategic

4) Pressure During Downturns

- In tough economic times, companies with high leverage may struggle to meet obligations, risking default or forced asset sales.

5) Reduced Flexibility

- High fixed costs (operating leverage) limit a company's ability to cut expenses during downturns.

1.13 TYPES OF LEVERAGE



1.14 OPERATING LEVERAGE

DEFINITION

Operating leverage shows the sensitivity of a company's operating income to changes in sales. The more fixed costs a company has relative to variable costs, the higher its operating leverage.

Operating leverage refers to the degree to which a company uses fixed costs in its cost structure. It measures how a change in sales volume affects operating income (EBIT).

Formula:

$$\text{DOL (degree of operating leverage)} = \frac{\% \text{ Change in EBIT}}{\% \text{ Change in Sales Or at a specific level of sales}}$$

$$\text{DOL} = \frac{\text{Contribution Margin}}{\text{Operating income (EBIT)}}$$

Operating Income (EBIT) Where:

- Contribution Margin = Sales – Variable Costs
- EBIT = Earnings Before Interest and Taxes

KEY CONCEPTS

➤ High Operating Leverage:

- High fixed costs, low variable costs.
- Small increases in sales lead to large increases in operating income.
- Riskier, but potentially more profitable when sales grow.
- Example industries: airlines, software, manufacturing.

➤ Low Operating Leverage:

- Low fixed costs, high variable costs.
- More stable profits, but less benefit from sales growth.
- Example industries: consulting, retail, food service.

FEATURES OF OPERATING LEVERAGE:

I. Involves Fixed Operating Costs

Operating leverage arises from the presence of fixed operating costs (like rent, salaries, and depreciation) in the cost structure.

II. No Involvement of Financial Costs

like financial leverage, it does not involve debt or interest payments—it focuses purely on business operations.

III. Magnifies Effect of Sales on EBIT

A small change in sales can lead to a larger change in operating income (EBIT) due to fixed costs.

IV. Measured by Degree of Operating Leverage (DOL)

$$DOL = \% \text{ Change in EBIT} / \% \text{ Change in Sales}$$
It quantifies how sensitive EBIT is to changes in revenue.

V. High Risk, High Reward

High operating leverage means higher business risk, but also higher potential returns when sales are growing.

VI. Break-Even Point is Higher

Companies with high operating leverage have a higher break-even point, as more sales are needed to cover fixed costs.

ADVANTAGES OF OPERATING LEVERAGE

- **Higher Profit Margins After Breakeven**

Once fixed costs are covered, additional sales directly boost profit, since variable costs are low.

- **Cost Efficiency**

Encourages the use of automation, technology, and efficient systems, reducing the need for labor - intensive variable costs.

- **Scalability**

Ideal for scaling businesses—revenues can grow faster than costs, increasing profit significantly.

- **Predictable Cost Structure**

Fixed costs make it easier to forecast and budget, improving operational planning.

DISADVANTAGES OF OPERATING LEVERAGE

- **High Breakeven Point**

A company needs to generate higher sales to cover fixed costs, increasing business risk.

- **Earnings Volatility**

A small drop in sales can cause a large decline in operating income (EBIT), due to fixed costs not decreasing.

- **Reduced Flexibility**

Fixed costs are hard to cut in tough times (e.g., rent, salaries, depreciation), limiting adaptability.

- **High Risk in Uncertain Markets**

In industries with unpredictable demand, high operating leverage can lead to significant losses.

1.15 FINANCIAL LEVERAGE:

DEFINITION

Financial leverage measures the degree to which a firm uses debt in its capital structure. It amplifies the potential return on equity, but also increases the risk if earnings do not cover interest payments.

Financial leverage refers to the use of borrowed funds (debt) to finance the acquisition of assets or operations, with the goal of increasing returns to equity shareholders.

FORMULA:

$$\text{Degree of Financial Leverage (DFL)} = \frac{\% \text{ Change in EPS}}{\% \text{ Change in EBI}}$$

Or at a specific EBIT level:

$$\text{DFL} = \frac{\text{EBIT}}{\text{EBIT} - \text{Interest Expense}}$$

KEY CONCEPTS

➤ **High Financial Leverage:**

- A small change in EBIT results in a larger change in EPS (Earnings Per Share).
- Increases potential returns but also financial risk.
- Common in capital-intensive industries (e.g., real estate, utilities).

➤ **Low Financial Leverage:**

- Lower debt obligations.
- More stable earnings, lower risk.
- Common in service-based or tech companies that rely more on equity.

FEATURES OF FINANCIAL LEVERAGE

1. **Use of Debt Capital**

Financial leverage involves the use of debt in the capital structure, as opposed to only relying on equity.

2. **Fixed Financial Cost**

Debt financing introduces fixed interest payments, which must be paid regardless of the company's earnings.

3. **Enhances Return on Equity (ROE)**

When the return on investment (ROI) exceeds the cost of debt, financial leverage can magnify returns for equity holders.

4. **Increases Financial Risk**

Higher leverage increases the risk of default and bankruptcy during downturns or periods of low earnings.

5. **Impact on Earnings Per Share (EPS)**

Financial leverage affects the EPS of a company. Positive leverage increases EPS, while negative leverage decreases it.

6. Optimal Capital Structure

Leverage plays a key role in determining the optimal mix of debt and equity to maximize firm value.

7. Degree of Leverage

Companies can measure leverage through metrics like the Degree of Financial Leverage (DFL) to assess its impact on earnings.

8. Irrelevance in Perfect Markets (Theoretically)

According to Modigliani and Miller's Proposition I, in a perfect market with no taxes or bankruptcy costs, financial leverage is irrelevant to firm value.

ADVANTAGES OF FINANCIAL LEVERAGE

- **Increased Return on Equity (ROE)**

When returns from investments exceed the cost of debt, financial leverage amplifies profits for shareholders.

- **Tax Benefits**

Interest payments on debt are tax-deductible, reducing the company's taxable income and overall tax burden.

- **Preserves Ownership**

Raising funds through debt avoids issuing new shares, so existing shareholders retain control of the company.

- **Lower Cost of Capital**

Debt is usually cheaper than equity, especially in low-interest-rate environments, helping reduce the company's weighted average cost of capital (WACC).

- **Access to Additional Capital**

Enables companies to undertake larger projects or expand faster than they could using only internal funds or equity.

DISADVANTAGES OF FINANCIAL LEVERAGE

1. Increased Financial Risk

Companies must repay debt with interest, regardless of business performance. This raises the risk of default or bankruptcy.

2. Earnings Volatility

While leverage can amplify profits, it can also magnify losses, making earnings per share (EPS) more volatile.

3. Debt Covenants and Restrictions

Lenders often impose covenants, which may limit business decisions (e.g., limits on further borrowing or dividends).

4. Lower Credit Rating

High debt levels can lead to downgrades in credit ratings, increasing the cost of future borrowing.

5. Reinvestment Risk

If the borrowed funds do not generate expected returns, the company is still obligated to repay, putting strain on cash flow.

1.16 COMBINED LEVERAGE:

DEFINITION

Combined leverage reflects how both:

Operating leverage (due to fixed operational costs), and Financial leverage (due to interest on debt) together influence the company's bottom line (EPS).

Combined leverage (also known as total leverage) measures the total risk a company faces by combining the effects of operating leverage and financial leverage. It shows how a change in sales will affect the company's earnings per share (EPS).

FORMULA:

Degree of Combined Leverage (DCL) = DOL×DFL Or:

$$DCL = \frac{\% \text{ Change in EPS}}{\% \text{ Change in Sales}}$$

At a specific level of operations:

$$DCL = \frac{\text{Contribution Margin}}{\text{EBIT} - \text{Interest Expense}}$$

KEY CONCEPTS:

➤ **High Combined Leverage:**

- A small change in sales leads to a large change in EPS.
- Indicates high business + financial risk.
- Suitable for companies confident in stable and growing revenue.

➤ **Low Combined Leverage:**

- More stable earnings.
- Lower potential return, but lower risk.

FEATURES OF COMBINED LEVERAGE:

1. Combination of Operating and Financial Leverage

It reflects the cumulative effect of fixed operating costs (like rent, salaries) and fixed financial costs (like interest on debt).

2. Measures Total Business Risk

Combined leverage helps assess total risk—both operating (business) risk and financial risk.

3. Indicates Sensitivity of EPS to Sales Changes

It shows how a small change in sales can lead to a much larger change in EPS (Earnings Per Share).

4. Measured by Degree of Combined Leverage (DCL)

$DCL = DOL \times DFL$ This formula multiplies the Degree of Operating Leverage and Degree of Financial Leverage to find total sensitivity.

5. Useful in Decision Making

Helps managers and investors evaluate the impact of sales fluctuations on profits and risk, guiding capital and operational decisions.

6. High Combined Leverage Means High Risk

7. Firms with high combined leverage face greater variability in EPS, making them riskier, especially during economic downturns.

8. Relevant for Capital-Intensive Firms with Debt

Companies with high fixed costs and significant debt are more likely to exhibit high combined leverage.

ADVANTAGES OF COMBINED LEVERAGE

1. High Profit Potential

When both sales and operations perform well, combined leverage can significantly boost Earnings Per Share (EPS) due to the double effect of fixed operating and financial costs.

2. Strategic Use of Resources

Companies can optimize their capital and cost structures, using a mix of fixed costs and debt to maximize profitability.

3. Accelerated Growth

Combined leverage allows for faster expansion by financing growth with debt while utilizing fixed costs for efficiency.

4. Tax Efficiency

Interest expenses reduce taxable income, offering tax benefits that improve net income.

5. Competitive Advantage

Efficiently leveraged firms can operate with lower per-unit costs and higher returns, potentially outperforming less-leveraged competitors.

DISADVANTAGES OF COMBINED LEVERAGE

1. High Risk Exposure

Combined leverage amplifies total risk—if sales decline, both fixed operating costs and interest obligations remain, leading to sharp drops in EPS.

2. Greater Earnings Volatility

Even minor changes in sales can cause large swings in net income, making earnings less predictable.

3. Reduced Financial Flexibility

Fixed commitments from both operating and financial leverage can limit a firm's ability to adapt to downturns or changing market conditions.

4. Higher Breakeven Sales Volume

A high level of fixed costs means the company must generate more sales just to break even, increasing vulnerability in slow markets.

5. Risk of Overleveraging

Mismanagement or overly aggressive use of leverage can lead to insolvency, especially during economic downturns or cash flow shortages.

1.17 EARNINGS PER SHARE:



ORIGIN OF EPS

➤ Early 20th Century – Birth of Corporate Reporting

- As stock markets expanded in the early 1900s, especially in the U.S. and U.K., there was a growing need for standardized performance metrics to help investors evaluate company profitability.
- Companies began reporting net income, and financial analysts started dividing this by the number of shares to estimate earnings per share, although it was not standardized.

➤ 1930s – EPS Gains Prominence in the U.S.

- The Securities Exchange Act of 1934 in the United States led to mandatory financial disclosures by publicly traded companies.
- Analysts and investors increasingly used EPS as a key profitability measure, even before it was formally defined.

➤ 1970s – Formal Standardization

- The Financial Accounting Standards Board (FASB) in the U.S. issued standards (notably FASB Statement No. 128) to standardize the calculation of EPS, distinguishing between:

✚ Basic EPS

✚ Diluted EPS

- Similar standards were adopted globally by other accounting bodies.

➤ Modern Accounting – International Standards

- International Accounting Standard (IAS) 33 – Earnings Per Share governs EPS reporting globally.
- It requires consistent presentation and calculation of both basic and diluted EPS in financial statements of listed companies.

1.18 DEFINITION OF EPS:

Earnings Per Share (EPS) is a financial ratio that indicates how much net profit is attributable to each outstanding share of a company's common stock.

1.19 MEANING OF EPS:

Earnings Per Share (EPS) represents the portion of a company's profit that is allocated to each individual share of common stock. It gives shareholders an idea of how much money the company is making per share they own.

1.19 TYPES OF EPS:

1. Basic EPS

- Definition: It is calculated using net income available to common shareholders divided by the weighted average number of common shares outstanding during the period.

Formula:

$$\text{Basic EPS} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Weighted Average Common Shares Outstanding}}$$

- Use: Most commonly reported in financial statements.

2. Diluted EPS

- Definition: It accounts for all potentially dilutive securities (like stock options, convertible bonds, etc.) that could increase the number of shares and reduce EPS.

Formula:

$$\text{Diluted EPS} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Weighted Average Shares} + \text{Dilutive Securities}}$$

- Use: Gives a more conservative estimate of earnings per share if all possible shares were issued.

1.20 ADVANTAGES OF EPS

1. Simple Indicator of Profitability

EPS offers a clear, per-share view of a company's profitability, making it easy to understand for investors.

2. Widely Used Performance Metric

It is a standard measure used by analysts, investors, and financial media to compare company performance.

3. Helps in Investment Decisions

EPS is a core input for valuation ratios like the Price-to-Earnings (P/E) ratio, aiding in stock selection and valuation.

4. Tracks Earnings Growth Over Time

EPS trends can indicate whether a company's earnings are growing, stable, or declining, helping assess long-term performance.

5. Useful for Comparing Companies

Especially within the same industry, EPS helps in benchmarking financial performance.

6. Linked to Shareholder Value

A rising EPS often correlates with increasing shareholder returns and may result in higher dividends or share prices.

1.21 DISADVANTAGES OF EPS

1. Affected by Accounting Policies

EPS can be manipulated through accounting practices like revenue recognition, depreciation methods, or asset revaluation.

2. Ignores Capital Structure Differences

EPS doesn't reflect how much debt or equity a company uses, which affects financial risk.

3. Not Comparable Across All Companies

Different share counts, preferred dividends, or business models can make EPS less useful for comparing firms across sectors.

4. Excludes Cash Flow Information

EPS is based on net income, not cash flow, so it may overstate earnings if profits are not backed by real cash.

5. Distorted by Share Buybacks

Companies can boost EPS artificially by repurchasing shares, reducing the denominator without improving actual performance

A STUDY ON FINANCIAL LEVERAGE AND ITS IMPACT ON EARNING PER

SHARE WITH REFERENCE TO MANDОВI MOTORS



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